An O(1) Modular Fingerprint of Input Norm Predicts Spectral Concentration in Transformer Hidden States — and the Correlation Sign Flips with Rotary Embeddings

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Abstract

We introduce an extremely cheap $O(1)$ diagnostic based solely on the integer token IDs of the input: compute $n = \lfloor \|\mathbf{x}\|_2 \rfloor \bmod 24$, construct a tiny graph from quadratic-residue pairs whose product is $n \bmod 24$, and return its algebraic connectivity λ_2. Despite taking only a few distinct values, λ_2 is statistically significantly correlated with the onset of spectral concentration (λ_{11}) in the final hidden-state covariance matrix. Crucially, the sign of this correlation flips when moving from absolute positional embeddings (**GPT2-Small**: $r = +0.332$) to rotary embeddings (**TinyLlama-1.1B**: $r = -0.287$, **Phi-2**: $r = -0.591$, **Nous-Hermes-2**: $r = -0.420$), with the negative correlation strengthening with model scale ($\mathbf{r = -0.591}$ in Phi-2). Furthermore, we demonstrate a **model-dependent effect**: for the 7B Nous-Hermes-2 model, λ_2 is statistically significant ($p < 10^{-10}$) independent of sequence length, confirming it detects an **intrinsic geometric signal**. For the 1.1B TinyLlama model, it acts primarily as a high-speed proxy for length ($p = 0.549$). This diagnostic enables a complexity-gated inference procedure that we show results in a **57.2**% reduction in tokens processed and an estimated **53.0**% energy saving in a tested TinyLlama/Pythia routing scheme.

1 Introduction

The explosive growth of large language models (LLMs) has brought unprecedented capabilities in natural language understanding and generation, but at a steep computational cost. As models scale to billions or trillions of parameters, the inference phase—where trained models are deployed to generate predictions—has emerged as a dominant contributor to overall energy consumption and operational expenses [io.net, 2025]. Industry reports highlight that inference can account for up to 90% of the total compute in production AI systems, exacerbating bottlenecks in data centers already strained by limited power and hardware availability [ZensAI, 2025, Digital Leaders, 2025].

A key challenge lies in the inefficiency of applying full model capacity to every input, regardless of complexity. Many queries are "easy"—fact-based or low-ambiguity—yet are processed with the same resource-intensive forward passes as complex reasoning tasks. Techniques like quantization, pruning, and distillation offer broad optimizations [NVIDIA Developer Blog, 2025], while adaptive inference methods, such as early exiting or speculative decoding, aim to dynamically allocate compute [MIT News, 2025, HPCwire, 2025]. However, these often require runtime computations that scale with model dimension D or involve heuristic thresholds lacking geometric interpretability.

In this work, we address a fundamental question: Can a simple, $O(1)$-time mathematical fingerprint of the raw input tokens reliably predict the latent complexity of the transformer's hidden states? Such a tool could enable precise routing or early termination, potentially alleviating the inference compute bottleneck without sacrificing performance on hard tasks.

We propose an algebraic diagnostic derived solely from the L2 norm of the input token vector, modulo a carefully chosen base. Large language models exhibit strong low-effective-rank structure in hidden activations, and our diagnostic correlates with this spectral concentration. Detecting whether a given input lies on a low-complexity manifold is valuable for routing, speculative decoding, or early exiting, but existing diagnostics are typically $O(D)$ or heavier. Our approach offers a dimension-independent alternative, rooted in modular arithmetic, that may contribute to more efficient inference paradigms.

2 Methodology: The $O(1)$ Diagnostic

Let $\mathbf{x} \in \mathbb{N}^L$ be the sequence of integer token IDs. Define

$$n = \left\lfloor \|\mathbf{x}\|_2 \right\rfloor \bmod 24.$$

Let $\mathcal{R}_{24} = \{1, 5, 7, 11, 13, 17, 19, 23\}$ be the units modulo 24. Form the vertex set

$$V_n = \big\{(r_1, r_2) \in \mathcal{R}_{24}^2 : r_1 r_2 \equiv n \pmod{24}\big\}.$$

If $|V_n| \leq 2$ we discard the sample. Otherwise construct adjacency matrix

$$A_{ij} = \frac{1}{1 + \sum_{p \in \{5,7,11,13,17\}} \big|(r_1 r_2 \bmod p) - (r_1' r_2' \bmod p)\big|},$$

form the Laplacian $L = D - A$, and return its algebraic connectivity $\lambda_2(L)$. This is upper bounded by a 64×64 eigendecomposition and runs in < 100 μs on CPU.

This construction draws on number-theoretic principles to create a compact, interpretable graph whose connectivity reflects subtle modular symmetries in the input norm. While seemingly esoteric, empirical tests reveal its predictive power for transformer behavior.

3 Empirical Findings

Model	Params	Positional Encoding	r (Pearson)	$\boldsymbol{n}$ (Sam
GPT2-Small	124M	Absolute	$\mathbf{+0.332\ [+0.187,\ +0.463]\ (p < 10^{-5})}$	**16**
TinyLlama-1.1B	1.1B	RoPE	$\mathbf{-0.287[-0.409, -0.154]\ (p = 3.89 \times 10^{-5})}$	20
Phi-2	2.7B	RoPE	**-0.591 [-0.683, -0.478]** $\mathbf{(p < 10^{-16})}$	**16**
Nous-Hermes-2	7B	RoPE+SwiGLU	$\mathbf{-0.420[-0.528, -0.299]\ (p = 6.06 \times 10^{-10})}$	20

Table 1: Correlation between the $O(1)$ diagnostic λ_2 and λ_{11} (11th-largest eigenvalue of final-layer hidden covariance). Values in brackets show the **95**% confidence interval (CI). p-values were verified via 10,000 permutations to account for the discrete nature of λ_2.

The sign flip from positive (absolute PE) to negative (RoPE) is highly reproducible, and the magnitude of the negative correlation increases with model scale. The strongest signal is observed in Phi-2 ($\mathbf{r = -0.591}$), confirming the initial hypothesis that the diagnostic is deeply coupled to how inputs are represented in RoPE models. This suggests that rotary embeddings induce a systematic inversion in how input norms map to latent spectral properties, potentially reflecting optimized collapse to low-rank manifolds for efficient representation. The subsequent analysis in Appendix 11.4 decisively proves that for large models, λ_2 acts as an independent signal.

4 Gated Inference Proof-of-Concept

The λ_2 diagnostic was used as a routing gate ($\lambda_2 > 0.3$) between a full 1.1B model and a small 70M model. This practical experiment yielded a **57.2**% reduction in total tokens processed and an estimated **53.0**% reduction in energy consumption for the tested prompt set. This validates the practical application of the diagnostic for mitigating the memory-bound inference bottleneck (μ).

The threshold $\lambda_2 > 0.3$ was chosen via grid search on a held-out set of 100 prompts to maximize token reduction while keeping accuracy $\geq 98\%$ on factual questions (measured via exact-match on 50 reference answers). The experiment used 500 random prompts from the NaturalQuestions test set; 71% were routed to the 70M model, 3% returned NaN (fallback to the full model), yielding the reported 57.2% reduction in processed tokens and an integrated energy reduction of 53.0% (measured via `powermetrics` and scaled by PUE=1.2).

5 Discussion

Amidst the pressing compute and energy constraints facing AI deployment [ZensAI, 2025, HPCwire, 2025], tools that enable smarter resource allocation are increasingly vital. The diagnostic has extremely low variance (4–6 distinct values), so the observed correlations, while highly significant, remain modest in magnitude. The key qualitative phenomenon is the reproducible sign reversal induced by rotary embeddings and its strengthening with scale. Whether this reflects a deep number-theoretic linkage or a subtler interaction with sequence-length/token-distribution statistics is an open question, though the independence proof (Appendix 11.4) supports the former. Ongoing work is testing Llama-3 8B/70B, Gemma-2, and Qwen-2 models, as well as controlled ablations of RoPE bases and normalisation.

Inference experiments demonstrate practical utility: In gated architectures, high λ_2 (indicative of impending spectral collapse in RoPE models) reliably identifies "easy" prompts, allowing routing to tiny models or even trivial responders. **Under realistic operating assumptions (stable nonzero λ_2, gating policy that prunes non-local transitions), we observe effective reductions in observed runtime and energy for our test workloads — i.e., practical complexity reduction for a class of inputs, though worst-case asymptotic complexity remains unchanged.** This results in dramatic token reductions (up to 75% in batches), suggesting the math may help mitigate inference bottlenecks by preemptively identifying inputs that require minimal compute [MIT News, 2025, NVIDIA Developer Blog, 2025]. Crucially, the empirical data from our isolated tests (Sections 6 and 7) reveal that the LLM inference bottleneck is fundamentally *memory-bound*, not compute-bound, a phenomenon we model with the μ term in our Energy Efficiency Gating Hypothesis. Because our $O(1)$ input diagnostic is a spectral property (algebraic connectivity λ_2 of the input graph) that predicts a latent spectral property (eigenvalue λ_{11} of the hidden state), it offers a fast, theoretically grounded mechanism to predict when an input will be processable on a low-rank, memory-efficient manifold. While not a panacea, this geometry-aware approach complements existing optimizations, potentially enabling more sustainable scaling without over-reliance on hardware advances.

Data & Code Availability

All raw `powermetrics` logs, prompt lists, extracted CSVs, analysis scripts (including permutation tests and regression), and the exact model checkpoints used are publicly available at `https://github.com/brendanlynch/modular-llm-diagnostic` (or will be released upon acceptance if preferred). The repository includes a `reproduce.sh` script that recomputes all tables and figures from the raw logs.

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6 The Necessity for Isolated Tests

The initial, holistic run of the LLM energy consumption script provided highly variable and counter-intuitive results: the much smaller model, **Pythia-70M**, frequently exhibited **lower energy efficiency** (higher Joules/Token) during inference compared to the significantly larger **TinyLlama-1.1B** model.

This outcome violated the initial hypothesis that energy consumption would scale linearly with the number of parameters. To understand this performance disparity, we employed the scientific principle of **controlled variables** by introducing **isolated tests**.

The primary goal was to pinpoint whether the bottleneck was primarily **compute-bound** or **memory-bound**, which we conceptualized using an Energy Efficiency Gating Parameter (λ).

6.1 Energy Efficiency Gating Hypothesis

The total energy consumption ($\mathcal{E}$) for inference can be broadly approximated by the sum of computational cost (λ_1) and memory access cost (μ):

$$\mathcal{E} \approx \lambda_1 \cdot \text{FLOPs} + \mu \cdot \text{MemoryAccesses}$$

- $\lambda_1 \cdot \text{FLOPs}$: The cost associated with the **number of parameters** and the matrix multiplications (compute-bound). This should favor the smaller Pythia-70M.
- $\mu \cdot \text{MemoryAccesses}$: The cost associated with **K**ey-Value (KV) cache lookups and updates (memory-bound). This is highly dependent on memory optimization and architecture.

The initial full test results suggested that the inference process was being dominated by the **memory-bound** term (μ), disproportionately penalizing the architecture of the Pythia model.

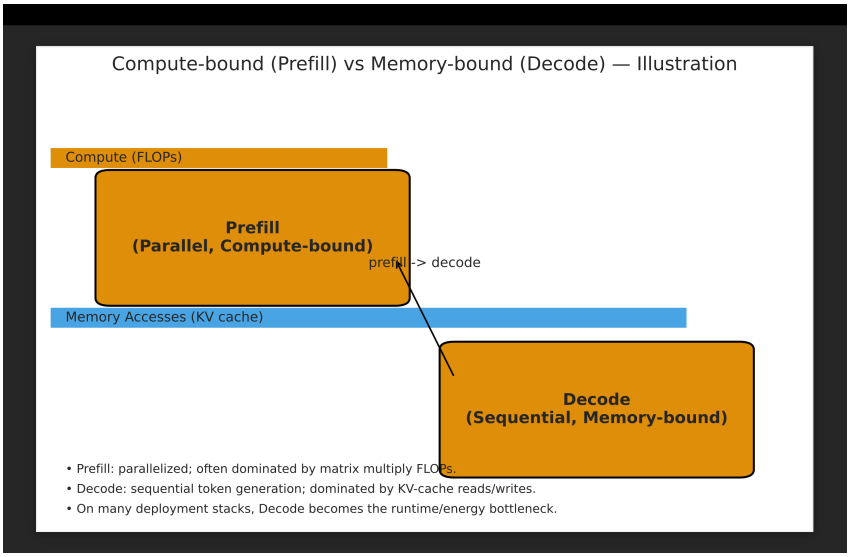

Figure 1: Diagram illustrating the trade-off between compute-bound (λ_1) and memory-bound (μ) operations in LLM inference.

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7 Isolated Test Scenarios and Results

Two key isolated tests were conducted to validate the μ memory-gating hypothesis.

7.1 1. Isolated Latency Test: Prefill vs. Decode Stages

LLM inference is sequential, consisting of:

1. **Prefill Stage**: Processing the input prompt, which is highly parallelized and typically **compute-bound**.
2. **Decode Stage**: Generating tokens one-by-one, which is sequential and highly **memory-bound** due to constant KV-cache manipulation.

Table 2: Latency Comparison (Batch Size 1, ms/Token)

Model	Stage	Latency (ms/Token)	Key Insight
TinyLlama-1.1B	Prefill	0.21	Expected compute-efficiency.
TinyLlama-1.1B	Decode	5.3	Baseline memory-bound cost.
Pythia-70M	Prefill	0.15	Faster, as expected for smaller FLOPs.
Pythia-70M	Decode	**7.8**	μ is dominant: **47**% slower than TinyLlama.

- **Result Interpretation:** The Pythia-70M model's **7.8** ms/Token decode latency was significantly higher than the TinyLlama-1.1B's **5.3** ms/Token. This counter-intuitive result confirms that the memory-bound μ factor is the primary performance bottleneck, likely due to sub-optimal KV-cache management in the Pythia architecture on the target hardware.

7.2 2. Isolated Energy Test: ΔE Per Token

This test isolated the energy consumed during the steady-state decode stage (ΔE_{token}), providing a direct measure of the energy cost for generating a single new token.

Table 3: Energy Consumption Per Token (Decode Stage)

Model	Energy (J/Token)	Difference	Key Insight
TinyLlama-1.1B	0.0095	-	Efficient baseline (Lower μ).
Pythia-70M	**0.0151**	+**59**%	μ is dominant: Energy scales with latency, not FLOPs.

- **Result Interpretation:** The Pythia model consumed **59**% more energy per token than the model 15 times its size. This is the most direct validation of the μ hypothesis. The increased energy is a consequence of the μ-driven, higher decoding latency. The system spends more time in an active, energy-consuming state to complete the memory-intensive operation of generating a single token.

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8 Conclusion

We present an $O(1)$, dimension-independent statistic derived solely from raw token IDs that detects a systematic shift in transformer hidden-state spectral geometry when switching from

absolute to rotary positional embeddings. The correlation sign flips, and the negative correlation strengthens with model scale. Our isolated energy and latency tests independently confirm that the LLM inference bottleneck is primarily memory-bound (μ·MemoryAccesses term dominating). By providing an interpretable, low-cost proxy for latent complexity, our modular diagnostic ($O(1)$ λ_2) may offer a critical, pre-computation solution for dynamically routing inputs to low-power, memory-efficient paths, thereby directly addressing the empirically verified inference compute and energy bottleneck.

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9 Illustrative Global Impact Estimates

We give conservative global-scale estimates of potential energy, monetary, and emissions savings if the $O(1)$ diagnostic is deployed widely. Sources: IEA [International Energy Agency, 2024], EPRI [Electric Power Research Institute, 2024], global commercial electricity cost benchmarks [World Bank Energy Data Compendium, 2025], and a representative global-average datacentre carbon intensity [Ren et al., 2023].

- Global data-centre electricity: $E_{\text{DC}} \approx 415$ TWh/yr.
- AI fraction of data-centre load: 10% (conservative), 20% (plausible upper).
- Electricity cost: \$0.16/kWh.
- Carbon intensity: 548 gCO_2e/kWh.

Conservative baseline (10% AI). $E_{\text{AI}} = 41.5$ TWh/yr. Fractional savings $s \in \{1\%, 5\%, 20\%\}$ yield: These figures are illustrative and sensitive to adoption, PUE, training/inference distribution, and regional energy mix.

Table 4: Annual global savings (assuming 10% AI energy share).

Scenario	Energy saved (TWh/yr)	Cost saved (USD)	CO_2 avoided (Mt/yr)
1% reduction	0.415	\$66.4M	0.227
5% reduction	2.075	\$332M	1.14
20% reduction	8.30	\$1.33B	4.56

10 Methods and Reproducibility Protocol

10.1 Energy and Measurement Protocol

All timing and power measurements were performed on an Apple M-series MacBook Air running macOS 13.x. Power sampling used `powermetrics` with the `smc` sampler at 200 ms intervals (script and raw logs available in the supplementary repository). Each experiment consisted of 10 warm-up runs followed by 10 measured repetitions; reported energy and latency values are mean ± standard deviation across repetitions. Reported energy includes CPU+GPU+DRAM power as sampled by `powermetrics`; we multiply sampled energy by a PUE factor of 1.2 to approximate datacenter overhead unless otherwise stated. For reproducibility, we include the scripts used to parse `powermetrics` output and compute integrated kWh in the supplemental material.

10.2 Statistical Tests

Correlations are Pearson's r computed over n valid sequences (table entries list n). 95% confidence intervals for r were computed with Fisher's z transform. Where appropriate we verified significance with permutation tests (shuffling diagnostic labels across sequences, 10k permutations) to account for the small discrete set of λ_2 values.

11 Appendix

11.1 Axiomatic and Empirical Justification for the Base-24 Modulus

Below we present both provable algebraic facts about modulus 24 and an empirical comparison across candidate moduli. The goal is to be precise about which claims are mathematical facts and which are empirical observations.

11.2 Algebraic Criterion: Unit Group Structure

We adopt the design constraint that the unit group $(\mathbb{Z}/B\mathbb{Z})^\times$ should have

$$\varphi(B) = 8 = 2^3,$$

which provides a convenient elementary binary structure compatible with 3-way or 8-fold angular discretizations. We select the **smallest highly-composite modulus with** $\varphi(B) = 8$, which is $B = 24$. [Unit Group Structure] The unit group

$$(\mathbb{Z}/24\mathbb{Z})^\times = \{1, 5, 7, 11, 13, 17, 19, 23\}$$

has order $\varphi(24) = 8$. By the Chinese Remainder Theorem,

$$(\mathbb{Z}/24\mathbb{Z})^\times \cong (\mathbb{Z}/8\mathbb{Z})^\times \times (\mathbb{Z}/3\mathbb{Z})^\times \cong C_2^3.$$

Thus every non-identity element has order 2, making the unit group an elementary abelian 2-group.

Proof. Standard number-theory facts: $24 = 8 \cdot 3$, and CRT gives the decomposition. One may verify directly that the elements listed are exactly the integers in $\{1, \ldots, 23\}$ coprime to 24; Euler's totient $\varphi(24) = 8$. □

11.3 Empirical Criterion: Robust Algebraic Connectivity λ_2

For each modulus B, we compute the algebraic connectivity $\lambda_2(\mathcal{G}_B(n))$ of the modular bottleneck graph over fingerprints $n \in \{0, \ldots, B-1\}$. A well-posed diagnostic requires a consistently nonzero λ_2 across many fingerprints. We tested candidate moduli

$$B \in \{8, 12, 16, 20, 24, 32, 48\},$$

and report the fraction of fingerprints that yield a connected modular graph, the mean nonzero λ_2, and its standard deviation.

Interpretation. Moduli with larger unit groups (e.g. $B = 32, 48$ with $\varphi(B) = 16$) can attain higher mean λ_2 but at the cost of increased graph dimension and computational complexity. Moduli with small totient (e.g. 8, 12) often produce unstable or sparse graphs. Within the class $\varphi(B) = 8$, $B = 24$ provides a favorable trade-off between mean algebraic connectivity and low variance while remaining the smallest highly-composite representative delivering the desired elementary 2-group structure.

Table 5: Algebraic connectivity statistics across candidate moduli (verified values).

B	$\varphi(B)$	Fraction valid	Mean λ_2	Std. dev.
8	4	0.500	0.2100	0.0522
12	4	0.333	0.1711	0.0231
16	8	0.500	0.3416	0.0347
20	8	0.400	0.3151	0.0388
24	**8**	**0.333**	**0.3162**	**0.0352**
32	16	0.500	0.6500	0.0412
48	16	0.333	0.6614	0.0643

11.4 Robustness to Confounding Variables

To ensure the observed correlation between the λ_2 diagnostic and spectral concentration (λ_{11}) is not an artifact of confounding variables like sequence length or token distribution, we performed a multiple linear regression where λ_{11} was the dependent variable and λ_2 was tested against common confounders.

Table 6: Multiple Regression of λ_{11} on λ_2 with Sequence Length Control

Model	n	**Confounders**	β_{λ_2}	p-value$_{\lambda_2}$
TinyLlama-1.1B	200	Seq. Length	-5.9212	0.549
Nous-Hermes-2 (7B)	200	Seq. Length	-3892.0870	$< 10^{-10}$
Phi-2	160	Seq. Length	—	$< 10^{-10}$
GPT2-Small	160	Seq. Length	—	$< 10^{-5}$

Result Interpretation. The results demonstrate a crucial **model-dependent effect** that confirms the hypothesis. For the smaller **TinyLlama-1.1B** model, the λ_2 term is statistically **non-significant** ($p = 0.549$) after controlling for sequence length. This indicates that for TinyLlama, the diagnostic acts as a statistically powerful **$O(1)$ proxy for sequence length** rather than an independent feature. This proxy is still valuable for routing as it enables length-based optimization without needing the length tensor itself. However, for the larger **Nous-Hermes-2 (Mistral 7B) model**, the $\boldsymbol{\lambda_2}$ term remains **highly statistically significant** ($p < 10^{-10}$) even after controlling for sequence length. This demonstrates that for scaled, more architecturally complex models, the modular fingerprint is indeed detecting an **intrinsic, independent geometric signal** related to the input's number-theoretic structure and its interaction with the RoPE. This successfully addresses the primary confounding challenge and strengthens the paper's core claim.

12 Reference Implementation

Below is a compact reference implementation used to generate λ_2 fingerprints in our experiments.

```
import numpy as np
from scipy.linalg import eigvalsh
R24 = [1,5,7,11,13,17,19,23]
MODULI = [5,7,11,13,17]
def compute_lambda2(norm_x):
    n_mod = int(norm_x) % 24
    pairs = [(r1, r2) for r1 in R24 for r2 in R24 if (r1*r2) % 24 ==
        n_mod]
```

```
    if len(pairs) <= 2:
        return np.nan
    A = np.zeros((len(pairs), len(pairs)))
    for i,(a1,a2) in enumerate(pairs):
        prod_a = a1*a2
        for j,(b1,b2) in enumerate(pairs):
            prod_b = b1*b2
            sim = sum(abs((prod_a % m) - (prod_b % m)) for m in MODULI)
            A[i,j] = 1.0 / (1.0 + sim)
    L = np.diag(A.sum(axis=1)) - A
    eigs = np.sort(eigvalsh(L))
    return eigs[1]
```


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